

Example 3: Determine if the following sets of vectors are linearly independent.

a) $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$ b) $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

Solution: a) Notice that v_2 is a multiple of v_1 , namely $v_2 = 2v_1$. Hence $-2v_1 + v_2 = 0$, which shows that $\{v_1, v_2\}$ is linearly dependent.

b) The vectors v_1 and v_2 are certainly not multiples of one another. Could they be linearly dependent? Suppose c and d satisfy

$$cv_1 + dv_2 = 0$$

If $c \neq 0$, then we can solve for v_1 in terms of v_2 , namely, $v_1 = (-d/c)v_2$. This result is impossible because v_1 is not a multiple of v_2 . So c must be zero. Similarly, d must also be zero. Thus $\{v_1, v_2\}$ is a linearly independent set. $\square$

A set of two vectors $\{v_1, v_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other. The set is linearly independent if and only if neither of the vectors is a multiple of the other.

Imp fact
In geometric terms, two vectors are linearly dependent if and only if they lie on the same line through the origin.

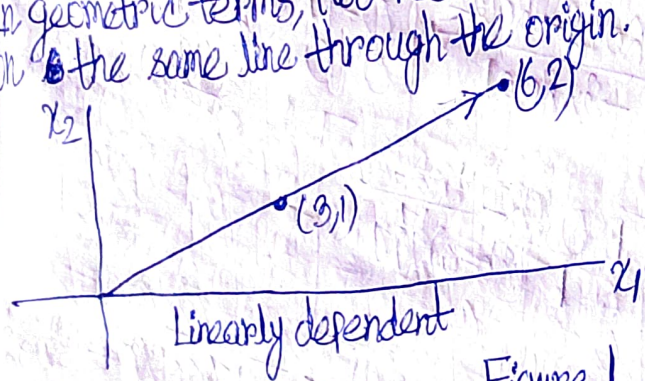
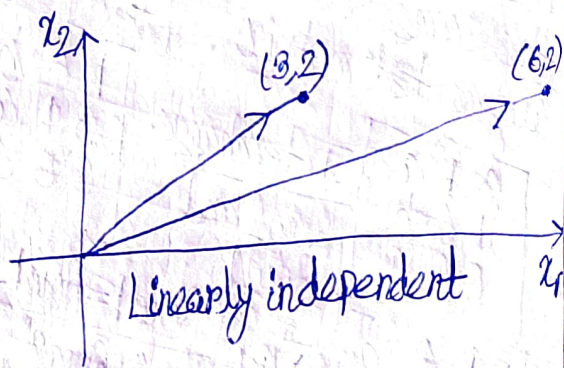


Figure 1



Sets of Two or More Vectors

Theorem 7: (Characterization of Linearly Dependent Sets)

An indexed set $S = \{v_1, \dots, v_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$,